



ROBO JANESH



TEAM



Arman Jamshidi

I'm Arman Jamshidi, 17 years old, and I have around 5 years of experience in robotics. This is my 2nd time participating in RoboCup. In our team, I've been responsible for Navigation, Movement and Mapping. Achievements:

- 1st place in Rescue Maze IranOpen 2026
- 2nd place in Rescue Maze IranOpen 2025
- 4th place in Rescue Simulation RoboCup 2024 Netherlands
- 3rd place in Rescue Maze Iran-Open 2024



Arya Torabi

I'm Arya Torabi. I have around 5 years of experience in robotics and this is my 2nd time participating. I've been responsible for Image Processing using Raspberry Pi 5. Achievements:

- 1st place in Rescue Maze IranOpen 2026
- 3rd place in Rescue Maze IranOpen 2025
- 4th place in Rescue Simulation RoboCup 2024 Netherlands
- 3rd place in Rescue Line IranOpen 2022



Sina Sabzevari

I'm Sina Sabzevari. I have around 5 years of experience in robotics and this is my 2nd time participating. I've been responsible for Mechanics and Electronics of robot. Achievements:

- 1st place in Rescue Maze IranOpen 2026
- 3rd place in Rescue Maze IranOpen 2025
- 4th place in Rescue Simulation RoboCup 2024
- 3rd place in Rescue Line Primary IranOpen 2023



Rayan Saeid Ershadi

I'm Rayan Saeid Ershadi, 17 years old, and I have around 3 years of experience in robotics. This is my 2nd time participating in IranOpen. In our team, I've been responsible for Navigation, Movement and Mapping. Achievements:

- 1st place in Rescue Maze IranOpen 2026
- 1st place in Rescue Simulation IranOpen 2024
- 4th place in Rescue Simulation Robocup 2024 Netherlands
- 2nd place in Rescue Simulation Iran-Open 2025
- 2nd place in Rescue Simulation 2025 Brazil



1x Lithium Ion Battery
It provides the robot with power, ensuring optimal performance for two complete rounds.



1x Raspberry Pi 5
Used as image processor with 8GB RAM, based on its impressive power as a single board computer.



Team Achievements



IranOpen 2026 - Iran, Tehran



IranOpen 2025 - Iran, Tehran



2nd place, Robocup 2026
Convention center in Salvador, Brazil

HARDWARE

The Programs

The robot body was designed and modeled using SolidWorks Designer software, which provided a highly flexible and accurate environment for creating the mechanical structure of the robot. During the design process, the primary objective was to make the robot as compact, lightweight, and efficient as possible while still maintaining the required strength and durability for rescue operations and challenging environments. A compact structure is one of the most important factors in the development of mobile rescue robots because it enables the robot to move through narrow passages, confined spaces, and difficult terrains that may be inaccessible to larger robotic systems.



Altium Designer software has been used for designing and developing the electronic circuits of this robot. The use of this professional PCB and electronic design software provides a reliable environment for creating accurate, organized, and efficient circuit layouts. The software allows the integration of schematic design, PCB layout, component management, and circuit analysis within a single platform, which significantly improves the efficiency of the design process. Through careful circuit organization and optimized routing, the electronic systems of the robot achieve greater stability, reliability, and electrical performance.



Chassis

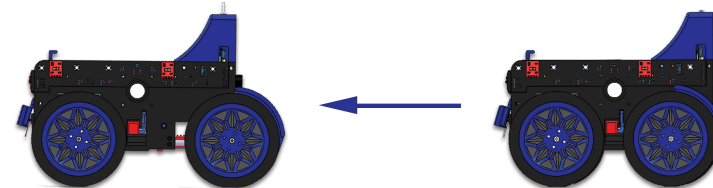
The robot was designed in SolidWorks and manufactured using custom 3D-printed PLA components. The chassis dimensions are 195 mm x 105 mm with a height of 95 mm, and a minimum wall thickness of 3 mm was used to improve durability while maintaining a lightweight structure. The mechanical design focuses on stability, modularity, and reliable operation in uneven rescue environments.

A key feature of the robot is its front-wheel suspension system, which allows all four wheels to maintain contact with the ground while traversing uneven surfaces. To prevent interference between moving components, the suspension travel is mechanically limited. This improves stability and traction during operation on ramps and obstacles.

To improve performance on ramps and speed bumps, the robot uses an adjustable wheelbase mechanism driven by a rack-and-pinion system. During ramp traversal, the distance between the wheel axes increases from 112.5 mm to 150.5 mm, improving balance and reducing the likelihood of tipping. Additional guide poles and ball bearings were incorporated to distribute loads and reduce stress on the mechanism.

The robot is powered by Dynamixel XC-430 actuators for locomotion and Dynamixel XL-330 actuators for rescue kit deployment and wheelbase adjustment. These actuators provide precise control, integrated feedback, and sufficient torque for reliable operation in competition conditions.

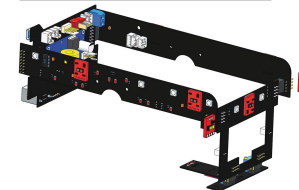
Axis Manoeuvre



To improve stability during ramp traversal and obstacle crossing, the robot uses a rack-and-pinion mechanism that dynamically increases the distance between the front and rear wheel axes. In normal operation, the wheelbase is 112.5 mm, while during ramp traversal it extends to 150.5 mm. This allows the robot to maintain better balance and distribute its weight more effectively on uneven terrain. Additional guide poles and ball bearings are used to support the mechanism and reduce mechanical stress on the moving components.

Stability is further enhanced by the robot's compact chassis, which measures 195 mm x 105 mm with a height of 95 mm. The low-profile design helps keep the center of gravity close to the ground, reducing the risk of tipping while traversing ramps, speed bumps, and other obstacles commonly found in rescue maze environments.

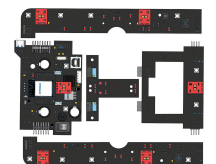
All Boards In One



The robot's electronics are implemented using a distributed multi-PCB architecture, where side boards are connected to the main board through pin-header interfaces. These connectors provide both power and signal communication between boards, enabling a modular structure that simplifies assembly, debugging, and replacement of components.

The side boards integrate key interfaces such as VL53L1X distance sensors, LEDs, and communication lines to other subsystems including motors and additional sensor boards. This structure reduces wiring complexity and improves signal reliability by minimizing loose connections and noise.

All sensor inputs, including color detection via the APDS-9960 sensor, are distributed across dedicated boards to improve organization and maintain clear separation between sensing and control layers. This design improves maintainability and allows individual sections of the electronics to be tested or replaced without affecting the entire system.



SOFTWARE

The Language

The control system code is written in C++ language and its programmed on ESP-32 while the Image processing is written with Python using different libraries, programmed and getting processed on Raspberry Pi 5.

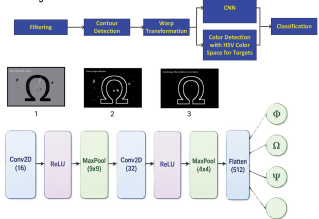


Image Processing

The robot's vision system runs on a Raspberry Pi 5 and follows a structured image-processing pipeline designed for efficient and reliable real-time performance. The processing flow includes sequential stages such as filtering, contour detection, geometric feature extraction, perspective warping, and final classification, where each step progressively refines the image and removes noise to improve detection accuracy. The final processed results are then transmitted to the ESP32 through serial communication for integration with the main control system, ensuring smooth coordination between perception and decision-making.

The robot's vision system uses two main intelligent detection approaches to improve reliability in competition environments. The first is a CNN-based letter recognition model, which processes cropped and warped images after preprocessing and contour extraction. The network automatically learns visual features through convolutional layers and performs final classification using fully connected layers, allowing it to distinguish between different victim letters with strong robustness to noise and lighting changes.

The second method is a cognitive target detection algorithm designed for circular patterns. It uses polar sampling to analyze pixel information in structured radial segments, and evaluates the results using a voting mechanism across sampled regions. This approach reduces the effect of local noise and improves stability under partial occlusion or uneven illumination.



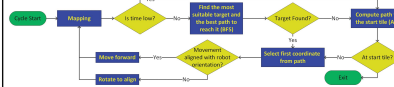
Control System

The robot's algorithm is built around a grid-based mapping system where each tile stores environmental information such as walls, floor color, and visited status. This allows the robot to maintain an internal representation of the maze and clearly distinguish between explored and unexplored areas during navigation.

A continuous decision-making cycle runs before each movement, where the robot updates its map using sensor data from distance sensors, compass, and floor sensors. Based on this updated map, it evaluates possible directions and selects the next target tile depending on the exploration state and overall navigation goals.

For path planning and exploration, the system uses a combination of A* and BFS algorithms. A* is used when an optimal path to a known target is required, while BFS is used during exploration to identify suitable next target tiles in unknown areas. This combination allows the robot to balance efficient goal-directed navigation with systematic exploration of the maze.

Together, the mapping system, decision-making cycle, and hybrid use of A* and BFS enable adaptive and structured navigation in complex and changing environments.



The robot's movement is based on tile-to-tile navigation, where each step corresponds to moving between discrete grid cells in the maze. After every movement, the robot updates its map and evaluates the next available directions based on the current environment, ensuring structured and predictable navigation in complex layouts.

During motion, encoder feedback and compass readings are continuously used to maintain accurate displacement and heading. If small errors occur, a PD-based correction adjusts the movement to keep the robot aligned with the intended path. Front and rear distance sensors are also used for fine alignment, helping the robot re-center before turns or transitions between tiles. Together, these methods ensure stable and reliable movement through intersections, narrow paths, and uneven maze conditions.